

BME 4313 "Tissue Mechanics"

Dr. Eric G. Meyer
Office: J-337
Voice: (248) 204-2606

Office hours: Tue/Thur 9:30 AM or
by appointment
E-mail:emeyer@ltu.edu

Class hours: 11-12:15 Tue/Thur in Engineering Building E100

Prerequisite(s): Undergraduate level BME 3033

Course Content

BME 4313 provides a more in-depth study of the mechanical properties and structural behavior of biological tissues and cells. Specific course topics will include the structure/composition and physiopathological relationships with the functional properties of tissues and cells; application of stress and strain analysis to biological tissues; derivation of viscoelasticity and nonlinear elasticity theories for tissues.

Course Instructional Format

The class will have lectures, in-class and out-of-class activities, and class discussion involving student participation. Students will be assessed based on quizzes, homework assignments, presentations and exams. This is a three credit class consisting of two 75 min lectures per week.

Required Textbook

Skeletal Tissue Mechanics (2nd Edition)

By; Martin, Burr, Sharkey, Fyhrie

ISBN 149393001

Recommended Textbooks

Biomechanics of Movement: The Science of Sports, Robotics and Rehabilitation

By; Uchida and Delp

ISBN 978-0-2620-4420-2

Basic Orthopaedic Biomechanics & Mechano-biology (3rd Edition)

By; Mow, Huiskes

ISBN 0-7817-3933-0

Introduction to Cell Mechanics and Mechanobiology

By; Jacobs, Huang, and Kwon

ISBN 978-0-8153-4425-4

Learning objectives

- Describe the material and structural properties of musculoskeletal tissues and explain similarities and differences between them.
- Explain how the constituents and organization of tissues contributes to mechanical behavior.
- Determine the internal forces acting on the human body during static and dynamic activities.
- Describe the stability and redundancy in the musculoskeletal system and use optimization techniques to solve for the muscle forces in a statically indeterminate problem.
- Formulate and solve for the quasi-linear viscoelastic and biphasic properties of biological tissues.
- Describe the techniques for determining material properties from experimental testing of biological tissues by using nonlinear and viscoelastic constitutive equations to fit these results.
- Explain how structural loading is transduced by the tissue and cells and how skeletal tissues adapt to mechanical loading.
- Describe basic structural components of a cell and explain how signal transduction pathways control the extracellular matrix regulation by the cells.
- Describe the history of biomechanics in relation to scientific development and identify key

- figures important to the historical and contemporary advancement of biomechanics.
- Explain the role of biomechanics in clinical methods for fracture repair, ligament repair, treatments for osteoporosis and osteoarthritis; including fixation devices, grafts, implants, and tissue engineering.
- Demonstrate the ability for self-directed learning by organizing and delivering effective written and verbal communication on a topic related to tissue mechanics or mechanobiology.

Grading

- **10% Attendance and Class Participation**

It is important that students attend class. Missing days will hurt your grade in this course. Please contact the instructor for any and all anticipated absences, and especially emergencies relating to class attendance. The student will get an "F" after the tenth unexcused absence.

- **5% Quizzes**

Weekly quizzes will be given throughout the course. Quizzes are participation based. One missed quiz will be excused, there will be no make-up quizzes.

- **20% Homework**

Students may discuss homework problems, but should write up solutions independently. Late submissions will not be accepted, unless previously arranged with the instructor.

- **10% Journal Club**

Student led presentations on a scientific journal article to summarize experimental methods, hypothesis driven research, and lead a discussion of the results and conclusions. All students are required to read the article and participate in the discussion for each journal club presentation. Additional details will be provided during the semester.

- **30% Midterm Exams**

Two equally weighted exams will assess the material in sections of the course. No substitute exams will be offered.

- **25% Final Exam**

A comprehensive final exam will assess the material in the whole course. No substitute exams will be offered.

Grading Scale:

A	> 93%
A-	90-92.9%
B+	87-89.9%
B	83-86.9%
B-	80-82.9%
C+	77-79.9%
C	73-76.9%
C-	70-72.9%
D+	67-69.9%
D	60-66.9%
F	<60%

For program assessment purposes, anonymous copies of submitted student work may be kept on file.

Attendance Rubric:

Participation Grade for Semester	Perfect Attendance	2 Absence	4 Absences	6 Absences	8 Absences	10+ Absences
A	10%	10%	9%	7%	5%	Fail
B	9%	9%	8%	6%	4%	Fail
C	8%	8%	7%	5%	3%	Fail
D	7%	7%	6%	4%	2%	Fail
F	5%	5%	4%	2%	0%	Fail

A = Fully engaged in class and team discussions. Asked and answered questions from instructor and classmates.

B = Mostly engaged in class and team discussions. Occasionally asked and answered questions from instructor and classmates. Occasionally distracted or engaged in other activities.

C = Only partially engaged in class discussions. Only responded when directly asked a question by the instructor. Occasionally distracted or engaged in other activities.

D = Rarely participated in class discussions. Only partially able to answer questions directly posed by professor. Frequently distracted or engaged in other activities.

F = Attended class but did not actively participate. Unable to answer questions directly posed by professor.

Course conduct

Students should read, understand and comply with the Lawrence Technological University Student Code of Conduct. For more information see:

http://www.ltu.edu/student_affairs/student_conduct.asp

Academic Integrity

Violations of academic ethics will be handled according to University policies as set forth in the University's Academic Honor Code see:

http://www.ltu.edu/currentstudents/honor_code.asp

Specific policies for assignments in this course are:

- All undergraduate students are required to write the following pledge on all academic work submitted along with a signature. If the pledge does not accompany an assignment, the faculty can choose to not grade it.
"I have neither given nor received unauthorized aid in completing this work, nor have I presented someone else's work as my own."
- Students may discuss assignments with others, but should complete their own work and write up their own solutions. All assignments and reports must be by individual or group effort only. Plagiarism, without appropriate reference citation, will not be tolerated.
- **Exams:** All exams must be by individual effort. Details will be provide on resources allowed to be used (calculators, laptops, notes, etc...) for each exam. Plagiarism will not be tolerated.

For program assessment purposes, anonymous copies of submitted student work may be kept on file.

Unforeseen circumstances may necessitate changes, additions, or deletions to this syllabus during the term.

Lawrence Technological University
BME 4313 “Tissue Mechanics”

Preliminary schedule of topics and assignments for the Spring 2026 semester.

Class	Date	Topic	Chapter	Assignments
1	1/13/26	LN1 Origins of Tissue Mechanics	Mow 1	
2	1/15/26	LN2 Structure of Tissues		JC0
3	1/20/26	LN3 Computing Joint Loads	Martin 1	HW1
4	1/22/26			
5	1/27/26	LN4 Computing Joint Loads cont.	Martin 1	JC1
6	1/29/26	LN5 Muscle Properties	Uchida 5	HW2
7	2/3/26	LN6 Muscle Optimization	Uchida 9	JC2
8	2/5/26	LN7 Ligament/Tendon Properties	Martin 8	Q2
9	2/10/26	LN8 Mechanics of Ligament/Tendon	Martin 8	JC3/HW3
10	2/12/26	LN9 Linear Viscoelasticity	Mow 5	Q3
11	2/17/26	LN10 Quasi-Linear Viscoelasticity	Mow 7	JC4
12	2/19/26	Exam 1 Review		
13	2/24/26	Exam 1		E1
14	2/26/26	LN11 Skeletal Development	Martin 2	HW4
15	3/3/26	LN12 Mechanics of Cartilage	Martin 7	JC5
16	3/5/26	LN13 Biphasic Behavior	Mow 5	Q4
17	3/17/26	LN14 Cartilage Regulation	Mow 6	JC6
18	3/19/26	LN15 Mechanics of Bone	Martin 4	HW5
19	3/24/26	Cont...		JC7
20	3/26/26	LN16 Bone Fracture	Martin 5	Q5
21	3/31/26	LN17 Implant Loading	Mow 14	JC8/HW6
22	4/2/26	LN18 Skeletal Adaptability	Martin 6	Q6
23	4/7/26	Cont...		JC9
24	4/9/26	Exam 2 Review		
25	4/14/26	Exam 2		E2
26	4/16/26	LN19 Cell Structure	Jacobs 2	HW7
27	4/21/26	LN20 Protein Mechanics	Jacobs 7	JC10
28	4/23/26	LN21 Cell Membrane	Jacobs 9	Q8
29	4/28/26	LN22 Membrane Mechanics	Jacobs 9	JC11
30	4/30/26	Final Exam Review		JC12
Final	TBD	Final Exam		FE